

Pranav Balabhadra

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Education

Case Western Reserve University

B.S. in Computer Science; Concentration in Artificial Intelligence

GPA: 4.00

Accolades: University Scholarship, Alumni Scholarship Recipient, National Merit Scholarship, Dean's High Honors List

Skills

Languages: Python, Kotlin, Typescript, C++, SQL

Tools: PyTorch, TensorFlow, Kubernetes, AWS, GCP, LangChain

Certifications: Fundamentals of Accelerated Computing with CUDA C/C++ (Issued by NVIDIA)

Work Experience

Software Engineer @ Amazon

Jun 2025 – Present

- Built and maintained Forge, an AI-powered developer tooling platform serving all client engineering teams across Prime Video, leveraging MCP to provide intelligent code search, documentation retrieval, and codebase understanding capabilities.
- Reduced Forge's Neptune Analytics costs by 93% (\$39k/month to \$2.8k/month) by designing a multi-tenant shared-graph architecture that consolidated 75+ Neptune graphs into just one.
- Designed and implemented an incremental indexing system for Forge's Neptune code intelligence graphs, replacing costly full rebuilds with targeted differential updates. Reduced graph construction time by 90% and shortened data sync cycles by 85%, dramatically improving freshness of code search results for engineering teams.
- Created a shared AI agent framework used by 50+ developers across the organization, providing specialized agents for code exploration, code review, and oncall triage with automated benchmarking to track agent quality over time.

Software Engineering Intern @ Iron Mountain

Jun 2024 – Aug 2024

- Implemented a CI/CD pipeline using ArgoCD, automating the deployment of an observability stack (Grafana, Prometheus, Loki) on Kubernetes, reducing deployment time by 80%.
- Implemented Kustomize to allow for environment-specific customizations, streamlining new deployments and reducing setup time by 70%.

AI Python Developer @ Expedient

Dec 2023 – May 2025

- Developed and maintained Expedient's AI CTRL services using React, TypeScript, and Python, enhancing user interaction and improving platform functionality.
- Incorporated support for various document formats using LangChain, enabling targeted questions related to provided content.
- Implemented a feedback system, allowing users to provide feedback on model responses, with data being sent to Elasticsearch for continuous improvement.

Software Engineering Intern @ Informetric Systems Inc

Jun 2023 – Aug 2023

- Implemented a high-performance program using C++ and OpenGL to enhance RQMS's graphing software, enabling the generation of intricate 3D surface plots with increased capabilities and significant improvement in compilation speeds.
- Configured and deployed an Amazon Elastic Compute Cloud instance with Amazon Relational Database Service to provide a proof-of-concept of AgileDoc in a cloud environment.
- Tested AgileDoc's functionality on the AWS infrastructure by verifying the deployment of software components including creating the SQL database and validating product functionality using SQL scripts and test scenarios.

Machine Learning Researcher @ Shoffstall Lab

Jan 2022 – May 2024

- Developed a robust machine learning model achieving a 92% testing accuracy for tracking fascicle movement in the vagus nerve from CT images.
- Integrated Meta's Segment Anything Model in conjunction with PyTorch into the image processing pipeline to improve segmentation accuracy.

Projects

MSE-Break: Steering Internal Representations to Bypass Refusals in Large Language Models (under review for ICLR 2026)

- Introduced MSE-Break, a soft-prompt-based method for steering harmful concept embeddings in LLMs to analyze and expose vulnerabilities in alignment mechanisms.
- Discovered that internal embeddings of harmful concepts are linearly separable between refused and accepted contexts, revealing structural weaknesses in safety-tuned models.

- Achieved 80–92% attack success rates, outperforming GCG (17–76%) and AutoDAN (27–65%), while converging 30x faster through gradient-based concept steering.
- Accepted to the Actionable Interpretability Workshop at ICML 2025: <https://tinyurl.com/msebreak>

Intelligent Document Processing Agent

- Developed an AI-powered pipeline for processing, extracting data from, and querying research papers in PDF format.
- Leveraged NLP techniques and Google's Gemini model to accurately extract complex features including images, figures, and tables from documents.
- Implemented a Retrieval-Augmented Generation model integrated with ChromaDB to index documents efficiently and retrieve contextually relevant content for answering user queries. Capable of answering complex queries within 8 seconds.